

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 How many electrons are needed to be removed to completely ionise 1.0×10^{-6} mol of K atoms to K^+ ions?

A 6.02×10^{23}
B 1.0×10^{-6}
C $1.0 \times 10^{-6} \times 6.02 \times 10^{23}$
D $\frac{1.0 \times 10^{-6}}{6.02 \times 10^{23}}$

- 2 0.200 mol of a hydrocarbon undergoes complete combustion to give 26.4 g of carbon dioxide and 10.8 g of water as the only products.

What is the molecular formula of the hydrocarbon?

A C_2H_4
B C_2H_6
C C_3H_6
D C_6H_{12}

- 3 A 50.0 cm^3 sample of an iron (II) solution required 34.60 cm^3 of $0.100 \text{ mol dm}^{-3}$ potassium dichromate (VI) solution for complete reaction.

Calculate the mass of iron (II) ions in 1 dm^3 of the iron (II) solution.

A 0.42 g
B 1.16 g
C 3.86 g
D 23.2 g

- 4 Elements **J** and **Q** have the following successive ionisation energies in kJ mol^{-1} .

	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th
J	587	1817	2745	11577	14842	18379	23326
Q	870	1790	2698	3610	5668	6820	13200

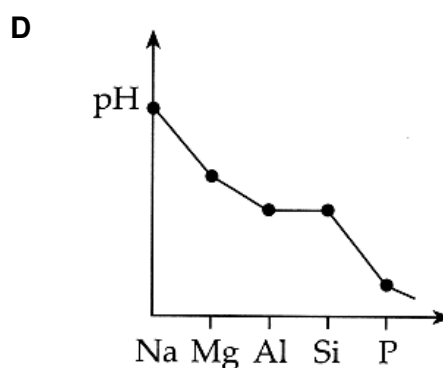
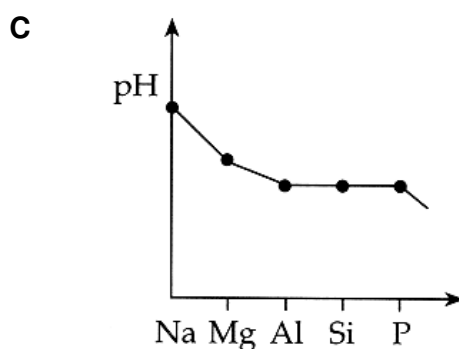
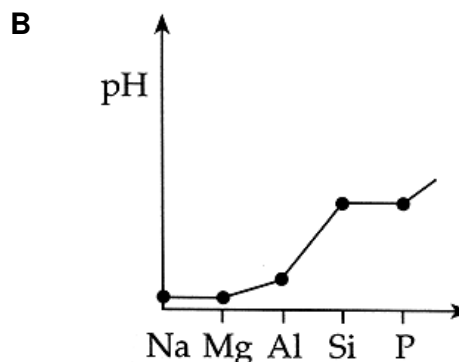
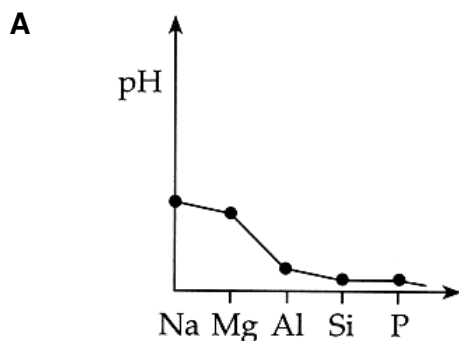
What is the likely formula of the compound that is formed when **J** reacts with **Q**?

A **JQ₃** **B** **J₂Q₃** **C** **J₃Q₂** **D** **J₂Q**

- 5 **X**, **Y** and **Z** are elements in Period 3 of the Periodic Table.
- **X** has the highest melting point among all the Period 3 elements.
 - The chloride of **Y** has the second highest melting point among all the Period 3 chlorides.
 - The oxide of **Z** is amphoteric.

Which of the following statements is **incorrect**?

- A** The atomic radius decreases in the following order: **Y** > **Z** > **X**.
- B** The electronegativity decreases in the following order: **X** > **Z** > **Y**.
- C** The electrical conductivity decreases in the following order: **Y** > **Z** > **X**.
- D** The first ionisation energy decreases in the following order: **X** > **Y** > **Z**.
- 6 One mole of the chloride of each of the following elements was mixed separately with 1 dm³ of water. Which one of the following graph shows the pH of the solutions produced?



- 7 In which of the following sequences are the species quoted in order of decreasing boiling points?

A Ti, K, Ca
B I₂, HF, HCl
C SCl₂, AlCl₃, MgCl₂
D CCl₄, CH₄, CH₂Cl₂

- 8 The shapes of three species **K**, **L** and **M** are trigonal pyramidal, bent and square planar respectively. Which of the following can be **K**, **L** and **M**?

	K	L	M
A	SO ₃ ²⁻	F ₂ O	XeF ₄
B	BCl ₃	SnCl ₂	ICl ₄ ⁺
C	PF ₃	CS ₂	BH ₄ ⁻
D	ICl ₃	ICl ₂ ⁻	BrF ₄ ⁻

- 9 The standard enthalpy change of formation of ethane is -84 kJ mol⁻¹.

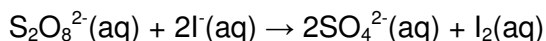
Given that the standard enthalpy change of combustion for carbon and ethane are -393 kJ mol⁻¹ and -1560 kJ mol⁻¹ respectively, what is the standard enthalpy change of combustion of hydrogen?

A -286 kJ mol⁻¹
B -417 kJ mol⁻¹
C +230 kJ mol⁻¹
D +361 kJ mol⁻¹

- 10 For which of the following is the lattice energy likely to have the greatest numerical value?

A lithium fluoride
B lithium iodide
C rubidium chloride
D sodium chloride

- 11 An experiment was carried out to investigate the initial rate of reaction between ammonia peroxodisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and potassium iodide, KI.



The initial concentrations of the $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and KI solutions in the mixture, together with the time taken for the mixture to darken for the various experimental runs are given below:

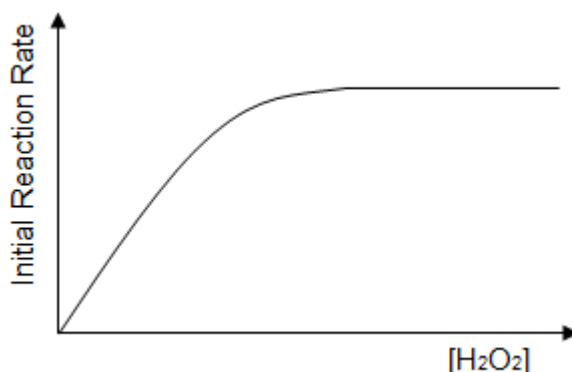
Expt	Initial concentration of $(\text{NH}_4)_2\text{S}_2\text{O}_8$ / mol dm^{-3}	Initial concentration of KI / mol dm^{-3}	Time taken to darken / s
1	0.10	0.20	35
2	0.05	0.20	70
3	0.10	0.067	105
4	0.02	0.75	?

Which of the following statements about the reaction is **correct**?

- A The rate constant has units of $\text{mol}^{-1} \text{dm}^{-3} \text{s}^{-1}$.
- B The reaction does not involve the formation of an intermediate.
- C The time taken for the mixture to darken in Experiment 4 is 80 s.
- D The slow step involves the reaction between 1 mole of $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and 1 mole of KI.
- 12 Which suggested mechanism is consistent with the experimentally obtained rate equation?

	rate equation	suggested mechanism
A	$\text{rate} = k_A[\text{H}_2][\text{I}_2]$	(slow) $\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$ (fast) $2\text{H}(\text{g}) + \text{I}_2(\text{l}) \rightarrow 2\text{HI}(\text{g})$
B	$\text{rate} = k_B[\text{HBr}][\text{O}_2]$	(slow) $2\text{HBr}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{HBrO}(\text{g})$ (fast) $\text{HBrO}(\text{g}) + \text{HBr}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + \text{Br}_2(\text{g})$
C	$\text{rate} = k_C[\text{NO}]^2[\text{H}_2]$	(slow) $2\text{NO}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g}) + \text{H}_2\text{O}(\text{g})$ (fast) $\text{N}_2\text{O}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
D	$\text{rate} = k_D[\text{NOBr}_2][\text{NO}]$	(fast) $\text{NO} + \text{Br}_2 \rightleftharpoons \text{NOBr}_2$ (slow) $\text{NOBr}_2 + \text{NO} \rightarrow 2\text{NOBr}$

- 13 The graph shows the results of an investigation of the initial rate of decomposition of hydrogen peroxide, H_2O_2 by the enzyme, catalase. The concentration of catalase was constant in all experiments.



Which one of the following statements is **false** about the above reaction?

- A When $[\text{H}_2\text{O}_2]$ is low, the rate is first order with respect to $[\text{H}_2\text{O}_2]$.
- B Increasing the amount of catalase will increase the rate of the reaction.
- C An increase in the system pressure will decrease the rate of decomposition of H_2O_2 .
- D At temperatures above 40°C , rate of reaction will decrease.
- 14 Two moles of compound **P** were placed in a vessel. The compound **P** was partly decomposed by heating. A dynamic equilibrium between chemicals **P**, **Q** and **R** was established. At equilibrium, x mol of **R** were present and the total number of moles present was $(2 + x)$. What is the equation for this equilibrium?

- A $\text{P} \rightleftharpoons 2\text{Q} + \text{R}$
- B $2\text{P} \rightleftharpoons 2\text{Q} + \text{R}$
- C $2\text{P} \rightleftharpoons \text{Q} + \text{R}$
- D $2\text{P} \rightleftharpoons \text{Q} + 2\text{R}$

- 15 Methanol is manufactured by reacting carbon dioxide and hydrogen.
- $$\text{CO}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g}) + \text{H}_2\text{O}(\text{g}) \quad \Delta H = -49 \text{ kJ mol}^{-1}$$
- What would increase the equilibrium yield of methanol in this process?

- A adding a catalyst
- B adding an excess of steam
- C increasing the pressure
- D increasing the temperature

- 16 Water dissociates as follows: $\text{H}_2\text{O}(l) \rightleftharpoons \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$. At a certain temperature above 298 K, the ionic product of water is $1.0 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$

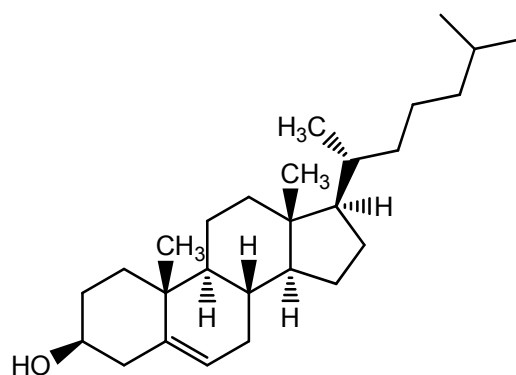
Which of the following deduction can be made from these information?

- A The total concentration of ions in the equilibrium mixture is $6.32 \times 10^{-7} \text{ mol dm}^{-3}$
 - B The water is acidic at this temperature.
 - C The ionic product of water is defined as $[\text{H}^+][\text{OH}^-] / [\text{H}_2\text{O}]$
 - D The dissociation of water is exothermic
- 17 Which of the following pairs of species **cannot** act as a buffer?
- A NaHCO_3 and H_2CO_3
 - B NaH_2PO_4 and Na_2HPO_4
 - C NaNO_3 and HNO_3
 - D CH_3COONa and CH_3COOH
- 18 Which formulae show propanone and propanal as different compounds?
- A Empirical, molecular, structural and displayed formulae
 - B Molecular, structural and displayed formulae only
 - C Structural and displayed formulae only
 - D Displayed formulae only
- 19 Acrylic acid is produced from propene, a gaseous product of oil refineries.

Which statement about acrylic acid is **not** correct?

- A Both bond angles x and y are approximately 120° .
- B It decolourises aqueous bromine.
- C It gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent.
- D It reacts with an alcohol to give an ester under appropriate conditions.

- 20 Which reaction would not give ethanoic acid as a product?
- A Heating ethanenitrile, CH_3CN , under reflux with dilute sodium hydroxide
 - B Heating ethanenitrile, CH_3CN , under reflux with dilute sulfuric acid
 - C Heating ethanal under reflux with acidified sodium dichromate(VI)
 - D Heating ethanol under reflux with acidified sodium dichromate(VI)
- 21 The diagram shows the structure of the naturally-occurring molecule cholesterol.



cholesterol

Student **X** claimed that the seventeen carbon atoms in the four rings all lie in the same plane.

Student **Y** claimed that this molecule displays cis-trans isomerism at the $\text{C}=\text{C}$ double bond.

Which of the students are correct?

- A both **X** and **Y**
- B neither **X** nor **Y**
- C **X** only
- D **Y** only

- 22** When 1-bromopropane is treated in succession with two reagents, **X** and **Y**, it produces propanoic acid. What are reagents **X** and **Y**?

	X	Y
A	NaOH(aq)	H ⁺ / Cr ₂ O ₇ ²⁻ (aq)
B	NaOH(aq)	CO ₂
C	KCN in ethanol	HCl(aq)
D	KCN in ethanol	NaOH(aq)

- 23** An organic compound **J** reacts with sodium to produce an organic ion with a charge of -3.

J reacts with NaOH(aq) to produce an organic ion with a charge of -1.

What could be the structural formula of **J**?

- A** HO₂CCH(OH)CH₂CO₂H
B HO₂CCH(OH)CH₂CHO
C HOCH₂CH(OH)CH₂CO₂H
D HOCH₂COCH₂CHO

- 24** How many isomeric esters have the molecular formula C₄H₈O₂?

- A** 2 **B** 3 **C** 4 **D** 5

- 25** CH₃COCH₂CH₂OH CH₃COCH₂CHO CH₃CH(OH)CH₂CH₃ CH₃CH₂CH₂CHO
 W **X** **Y** **Z**

Which of these compounds can be oxidised by acidified dichromate(VI) solution and also gives a positive response to Tollens' reagent?

- A** **W** and **X** only
B **W** and **Y** only
C **X** and **Z** only
D **Y** and **Z** only

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

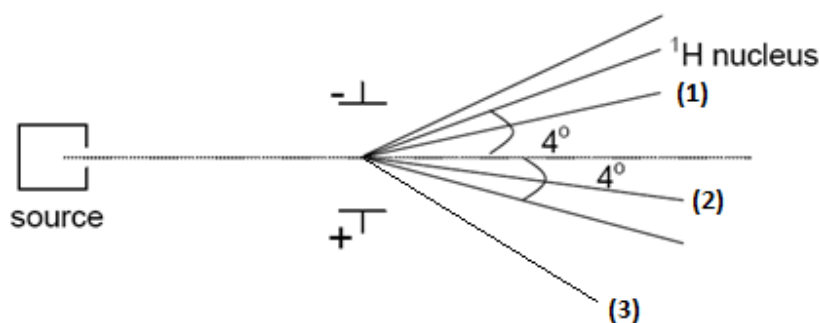
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** A stream which contains a mixture of gaseous particles was passed through an electric field. One of the particles, ${}^1\text{H}^+$ was deflected as follows. Which of the following particle(s) is/are correctly represented by the different line(s)?



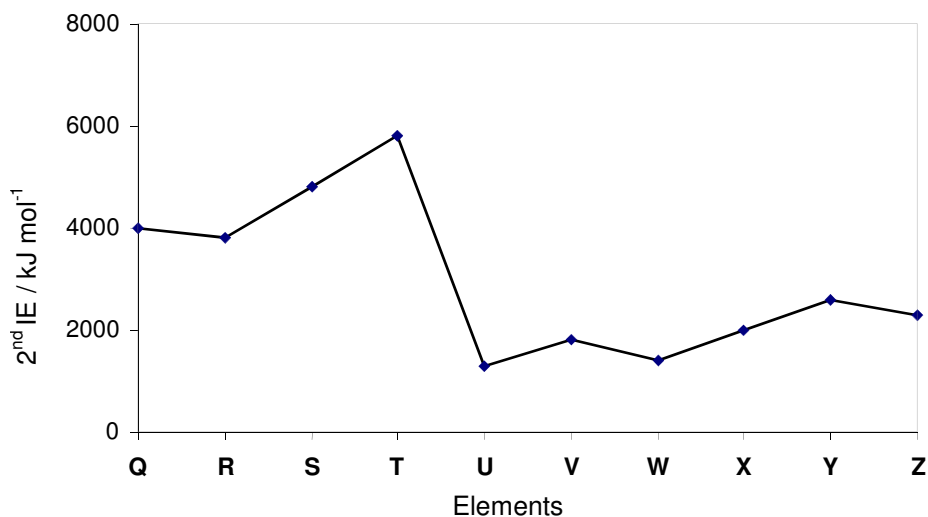
- 1** Line (1): ${}^4\text{He}^{2+}$
- 2** Line (2): ${}_{-1}^0\text{e}$
- 3** Line (3): ${}^{16}\text{O}^{2-}$

- 27** A refractory material is one that retains its strength at high temperature. These materials have chemical and physical properties which make them applicable for structures which are exposed to environments with temperatures higher than 2000°C .

Which of the following compound(s) can be used as a refractory material?

- 1** Silicon carbide
- 2** Magnesium oxide
- 3** Gallium trichloride

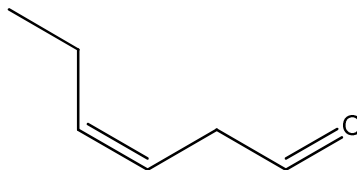
- 28 The graph below shows the variation in the **second** ionisation energies for the consecutive elements **Q** to **Z** in the Periodic Table, all with proton number below 20.



What can be deduced from the above?

- 1 **Q** has a smaller ionic radius than **R**.
 - 2 Effervescence is observed when a magnesium strip is dipped into the aqueous solution containing the chloride of element **V**.
 - 3 The oxide of **T** dissolves in water to form a alkaline solution.
- 29 Which statements about the photochemical chlorination of ethane are correct?
- 1 Hydrogen gas is one of the products.
 - 2 A propagation step in the mechanism is $\text{C}_2\text{H}_6 + \text{Cl} \cdot \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$.
 - 3 The initiation step is the homolytic fission of chlorine.

- 30 The compound *cis*-hex-3-enal is responsible for the characteristic smell of cut grass. The human nose is particularly sensitive to this compound, being able to detect 0.25 parts per billion in air.



cis-hex-3-enal

Which reagents will react with *cis*-hex-3-enal?

- 1 sodium
- 2 sodium borohydride
- 3 Fehling's reagent

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